Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Driving force

Meaning in my words How would you explain driving force to a friend?	**How can you tell?** A rope pulls right while friction acts left. Which one is the driving force on the cart?
Example or sketch Draw a 2 N driving pull and 0.4 N of friction. Which arrow is the driving force?	**Non-example and why** Is the hanging weight always the same as the rope tension on the cart?

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Slope

Meaning in my words How would you explain slope to a friend?	**How can you tell?** Which cart has less mass if its acceleration rises more per newton?
Example or sketch Net force rises 4 N and acceleration rises 2 m/s². What mass fits that slope?	**Non-example and why** A partner reads the acceleration at one point and calls it slope. What two changes do you need?

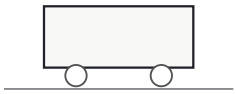
Friction offset

Meaning in my words How would you explain friction offset to a friend?	**How can you tell?** If the pull is 3 N and opposing friction is 1 N, how much force accelerates the cart?
Example or sketch For a sliding 2 kg cart with 1 N opposing friction, what acceleration would a 3 N pull produce?	**Non-example and why** A student uses the full 3 N pull as the net force. What did they leave out?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the pull and opposing friction. Label driving force on one arrow and net force on a separate arrow.



My marks mean

Which run changes more?

12

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when net forces differ.

CHOOSE ONE JOB TOGETHER

FLIGHT-TEST ENGINEER ☐

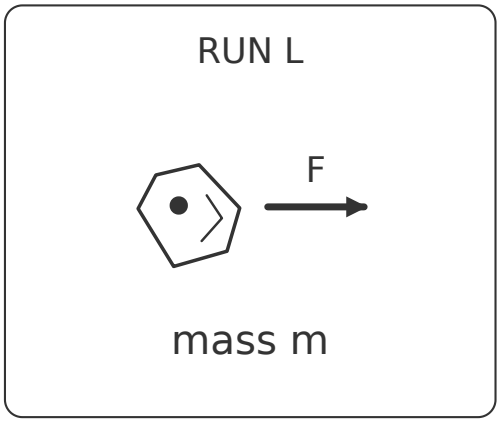
Add velocity-change arrows for both runs. Recommend the larger change and name what stayed the same.

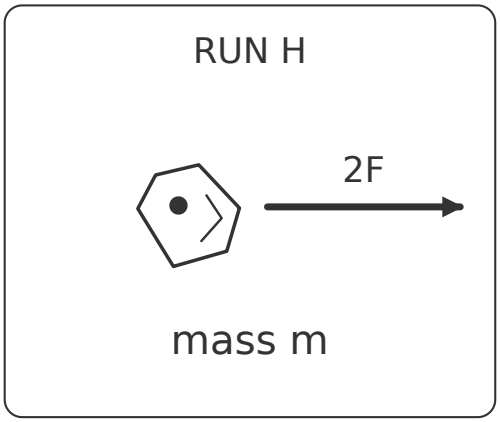
TEST-RANGE BROADCASTER ☐

Tell viewers which run changes velocity more. Ask whether doubling motor pull always doubles the net force.

Point to a clue. Keep your first marks; add the update using the color key.

A rehearsal uses the same asteroid model twice. The second net push is twice the first, for the same time. Operations needs a comparison.





01 OUR FIRST CALL

How do the two velocity changes compare? Point to the force arrows.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Do the new readings still support your comparison? Explain what you must check.

Switch speakers. Explain the difference between doubling one applied force and doubling the net force.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks